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DUNMAN HIGH SCHOOL
Preliminary Examination
Year 6

H2 BIOLOGY

9648/01

Paper 1 Multiple Choice Questions

30 September 2014

1 hour 15 mins

Additional Material: OTAS sheet

INSTRUCTIONS TO CANDIDATES:

DO NOT TURN THIS PAGE OVER UNTIL YOU ARE TOLD TO DO SO.

READ THESE NOTES CAREFULLY.

Section A MCQ [40 marks]

There are **forty** questions in this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

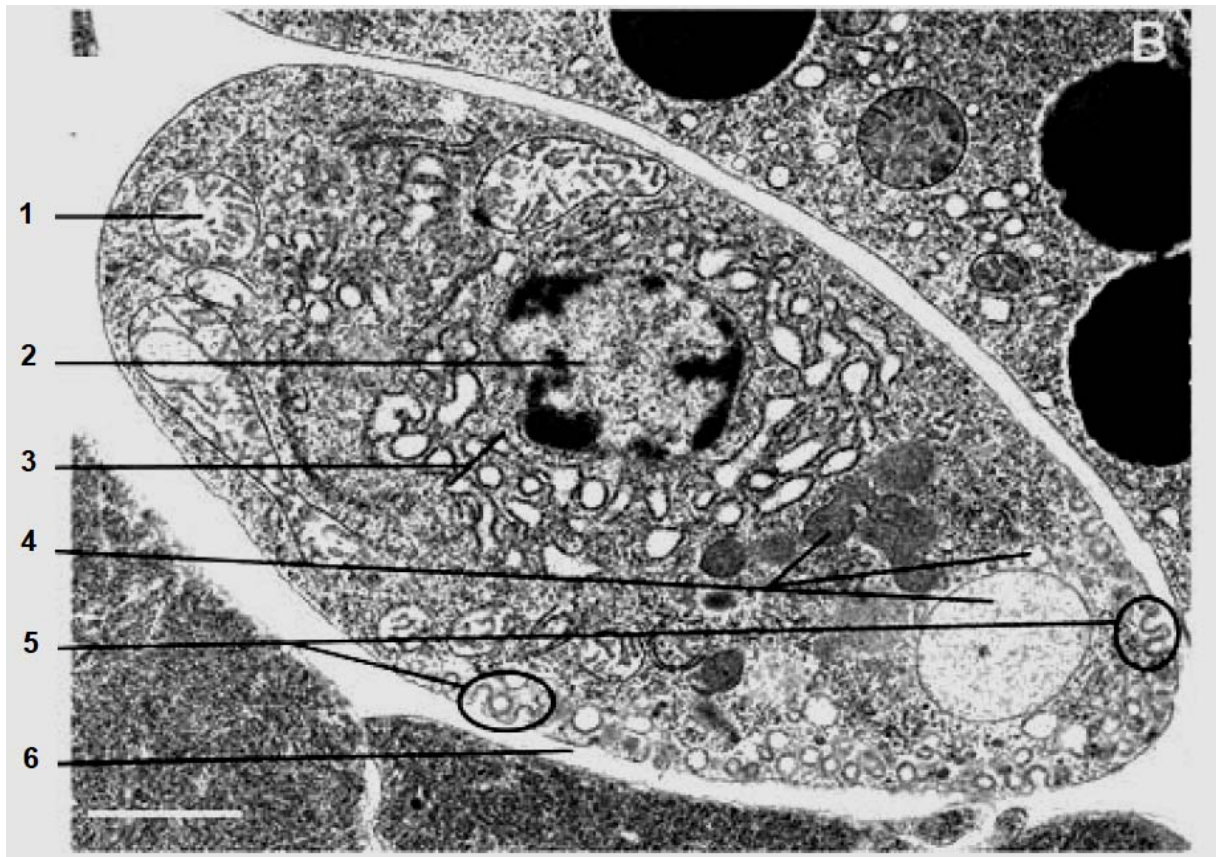
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

This document consists of **29** printed pages and **1** blank page.

Answer **all** questions in this section.

1



Which of the structure(s) in the above electron micrograph is/are found abundantly in actively respiring tissues?

- A 1 only
- B 4 only
- C 2 and 5 only
- D 3 and 6 only

2 Which of the following organelles are DNA present?

- A Chloroplasts and ribosomes
- B Chloroplasts and mitochondria
- C Mitochondria and lysosomes
- D Nucleoli and ribosomes

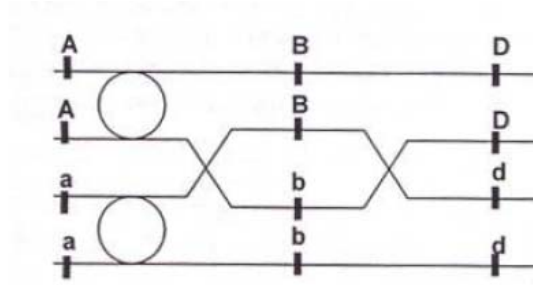
- 3 Some germinating seeds were crushed with water and the extract was tested. The table below shows the results of these tests.

Test	Result
Barford's test for disaccharides	Positive
Benedict's test	Yellow precipitate
Biuret test	Purple colour
Clinistix test for glucose	Negative
Emulsion test	Clear solution
Iodine test	Blue-black colour

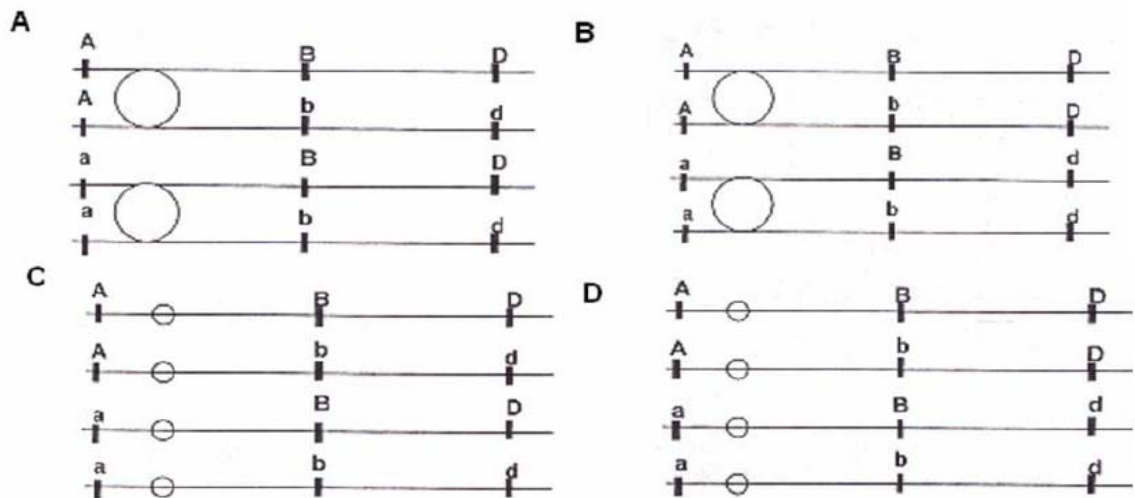
Which molecules were present in the extract?

- A Fat, maltose and starch only.
- B Glucose, protein and starch only.
- C Maltose, protein and starch only.
- D Sucrose, protein and starch only.

4



The diagram above shows a bivalent. Which of the following correctly represents the final products at the end of Meiosis II?



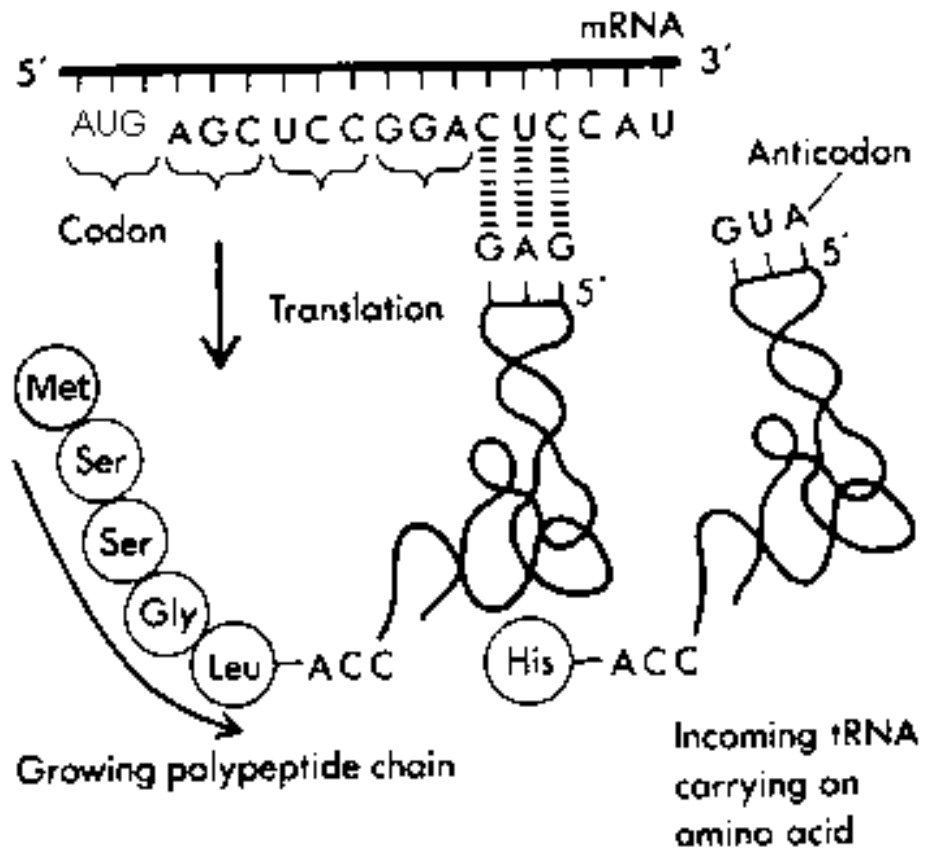
- 5 An experiment is carried out to show that DNA replication is semi-conservative.

Escherichia coli cells whose DNA contains only 'heavy' nitrogen (^{15}N) are grown in a culture medium containing the normal isotope of nitrogen (^{14}N) and allowed to divide three times.

What will be the percentage of DNA molecules containing both forms of nitrogen after the third division?

- A 12.5%
- B 25%
- C 50%
- D 100%

- 6 The diagram below shows the process of translation.



Which of the following are **CORRECT**?

- I The ribosome is translocating from right to left.
 - II The diagram shows degeneracy of the genetic code.
 - III The polypeptide would be attached to the tRNA after histidine was added.
 - IV The number of hydrogen bonds formed between the respective codons and anticodons of the two tRNAs are equal.
- A I and IV
 B I and II
 C II and III
 D II and IV

7 What is the basis for the difference in the synthesis of the leading and lagging strands of DNA molecules?

- I** The anti-parallel arrangement of the DNA strands.
- II** The RNA primers are required to initiate DNA elongation.
- III** The template strands must be read from the 3' to 5' direction
- IV** Helicase and single-stranded binding proteins work at the 5' end of the DNA strand.

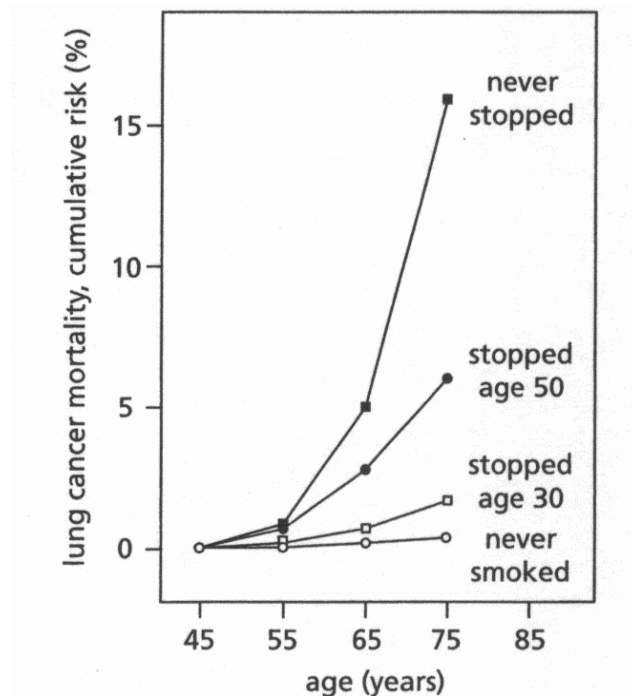
A I and III

B I and IV

C II and III

D II and IV

- 8 Mortality due to lung cancer was followed in groups of males in the United Kingdom for 50 years. The cumulative risk of dying from lung cancer as a function of age and smoking habits for four groups of males is shown in the figure below.



Which of the following best explains the trends observed in the graph?

- I The slower the rate of accumulation of mutations, the lower the cumulative risk.
 - II When an individual stops smoking, he will undergo a decreased rate of mutation accumulation.
 - III The presence of cigarette smoke will cause the accumulation of mutations to occur at an increased rate.
 - IV There is little risk of a non-smoker dying of lung cancer.
- A I and IV only
- B II and III only
- C III and IV only
- D All of the above

9 Some events that take place during generalised transduction are listed below.

- 1 Bacterial host DNA is fragmented
- 2 Bacterial DNA may be packaged in a phage capsid
- 3 Recombination between donor bacterial DNA and recipient bacterial DNA
- 4 Phage infects a bacterial cell
- 5 Phage DNA and proteins are made

Which sequence of events is **CORRECT**?

- A** 4, 1, 3, 5, 2
 - B** 4, 1, 5, 2, 3
 - C** 4, 3, 1, 5, 2
 - D** 4, 5, 1, 3, 2
- 10** Scientists have discovered how to put together a bacteriophage with the protein coat of phage T3 and the DNA of phage T4. If this composite phage were allowed to infect a bacterium, the phages produced in the host cell would have _____.
- A** the protein coat of T3 and the DNA of T4
 - B** the protein coat and DNA of T3
 - C** the protein coat of T4 and the DNA of T3
 - D** the protein coat and DNA of T4

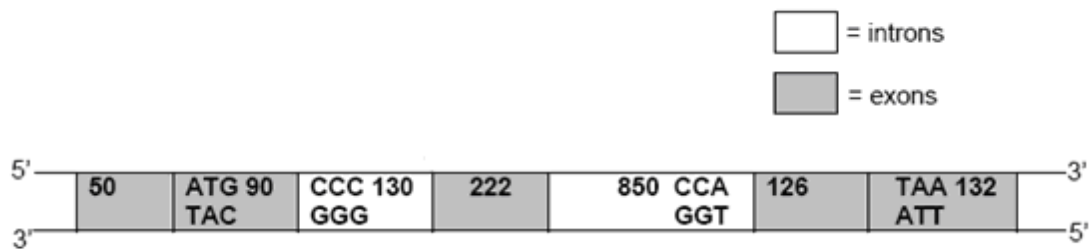
11 The following statements describe bacterial conjugation.

- I The F plasmid is made of single-stranded DNA.
- II When an F⁺ donor gives an F plasmid to an F⁻ recipient, both become F⁻.
- III When an F⁺ donor gives an F plasmid to an F⁻ recipient, both become F⁺.
- IV When an F⁺ donor gives an F plasmid to an F⁻ recipient, the donor becomes F⁻.

Which of the statements is/are **TRUE**?

- A III only
- B I and III
- C III and IV
- D I, II and IV

12 The diagram below shows the genomic structure of a human gene. The numbers within the boxes indicate the length of nucleotides of each region, inclusive of the bases stated in the diagram. The DNA sequences corresponding to the start codon and the stop codon are also indicated.



What is the length (in nucleotides) of the primary RNA transcript (pre-mRNA) and how many amino acids are present in the protein?

	Length of primary RNA transcript	Number of amino acid in protein
A	1 600	146
B	620	146
C	1 600	206
D	620	206

13 The following events occur in the extension of telomeres.

- 1 Further extension of 3' end of telomere.
- 2 Translocation of telomerase to end of telomere.
- 3 Reverse transcription to extend the 3' end of the telomere.
- 4 Complementary base-pairing of RNA template with single-stranded end of telomere.
- 5 DNA polymerase catalyses formation of complementary strand to form double-stranded DNA.

In which order does these events take place?

- A** 3 2 4 5 1
- B** 3 4 2 1 5
- C** 4 2 3 5 1
- D** 4 3 2 1 5

14 Cells taken from a human bone cancer multiplied readily in culture. Analysis showed that the cells were unable to produce the protein, RB.

Addition of RB to these cells reduced their rate of division.

Which of the following statement accounts for the observation?

- A** Both chromosomes in the cancer cell carry alleles for tumour suppressor gene.
- B** Both chromosomes in the cancer cell have the allele for tumour suppressor gene deleted.
- C** Both chromosomes in the cancer cell carry alleles for proto-oncogene.
- D** Both chromosomes in the cancer cell have the allele for tumour suppressor gene deleted.

- 15** Ribonuclease is an enzyme that digests RNA. The first five amino acids of the functioning molecule of ribonuclease are

lys – glu – thr – ala – ala

The mRNA of the gene coding for ribonuclease, for the first 15 nucleotides, has the following sequence.

AUGAAGGAAACUGCU

A genetic code, showing mRNA codons, is shown below.

first position	second position				third position
	U	C	A	G	
U	phe phe leu leu	ser ser ser ser	tyr tyr STOP STOP	cys cys STOP trp	U C A G
C	leu leu leu leu	pro pro pro pro	his his gln gln	arg arg arg arg	U C A G
A	ile ile ile met	thr thr thr thr	asn asn lys lys	ser ser arg arg	U C A G
G	val val val val	ala ala ala ala	asp asp glu glu	gly gly gly gly	U C A G

Which events occurs to explain the information given above?

- A** There is no tRNA with an anticodon complementary to the first codon.
- B** The first codon is removed from the mRNA transcript in post-transcriptional modification.
- C** The mRNA binds to the rRNA in the second codon position.
- D** The first amino acid on polypeptide chain is removed in post-translational modification.

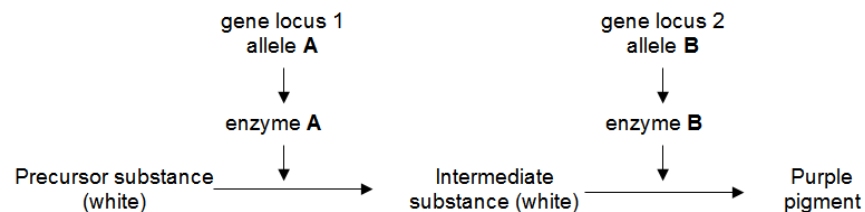
- 16 The table shows the blood group phenotypes resulting from crosses between different genotypes.

Genotype of 1 st parent	Genotype of 2 nd parent	
	I ^A I ^A	I ^A I ^O
I ^A I ^O	A	A and O
I ^B I ^O	A and AB	A, B, O and AB

Two parents have a son who has blood group A and phenylketonuria. One parent has blood group O and the other has blood group AB. Neither parent has phenylketonuria. What is the probability that the second child of these parents will be a girl with blood group B who does not have phenylketonuria?

- A 1 in 16
- B 1 in 8
- C 3 in 16
- D 3 in 8
- 17 The inheritance of colour in sweat peas is controlled by two pairs of alleles (**A** and **a**, and **B** and **b**) at two different gene loci.

The flowchart below shows how the dominant alleles at these two gene loci determine flower colour by controlling the synthesis of a purple pigment.

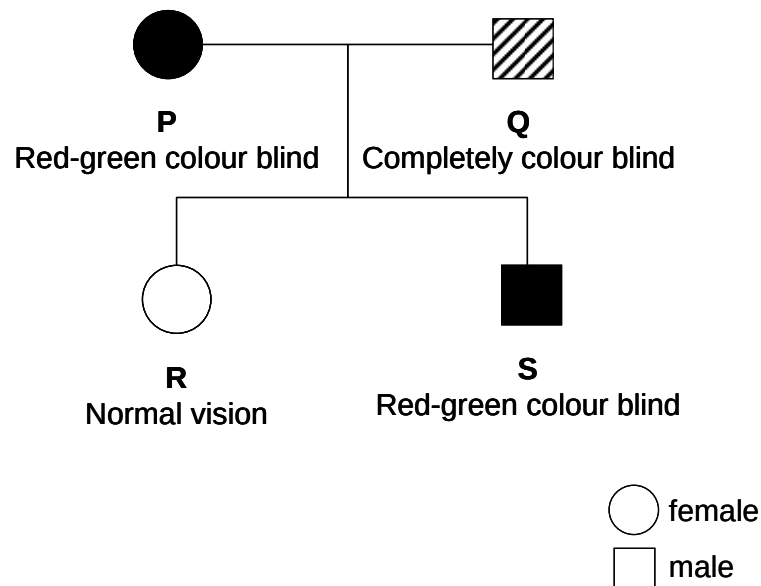


If a double heterozygote plant (**AaBb**) is *self-pollinated*, what is the percentage of phenotypes observed in the resulting offspring that are white?

- A 7%
- B 19%
- C 44%
- D 56%

- 18** Red-green colour blindness is controlled by a gene on the X chromosome. The allele for colour blindness, **g**, is recessive to the allele for normal colour vision, **G**. Complete colour blindness is controlled by a different gene which is not on the X chromosome. The allele for the development of normal cones (pigment cells in the retinal layer of the eye), **B**, is dominant to the allele for no cone development, **b**.

The figure below shows the phenotypes of members of a different family in which both types of colour blindness occur.



Which of the following are possible genotypes for individuals **P**, **Q** and **S**?

	P	Q	S
A	$BBX^G X^G$	$bbX^G Y$	$BBX^g Y$
B	$BbX^G X^g$	$BbX^g Y$	$BbX^G Y$
C	$BBX^g X^g$	$bbX^G Y$	$BbX^g Y$
D	$BbX^G X^g$	$BbX^g Y$	$BBX^G Y$

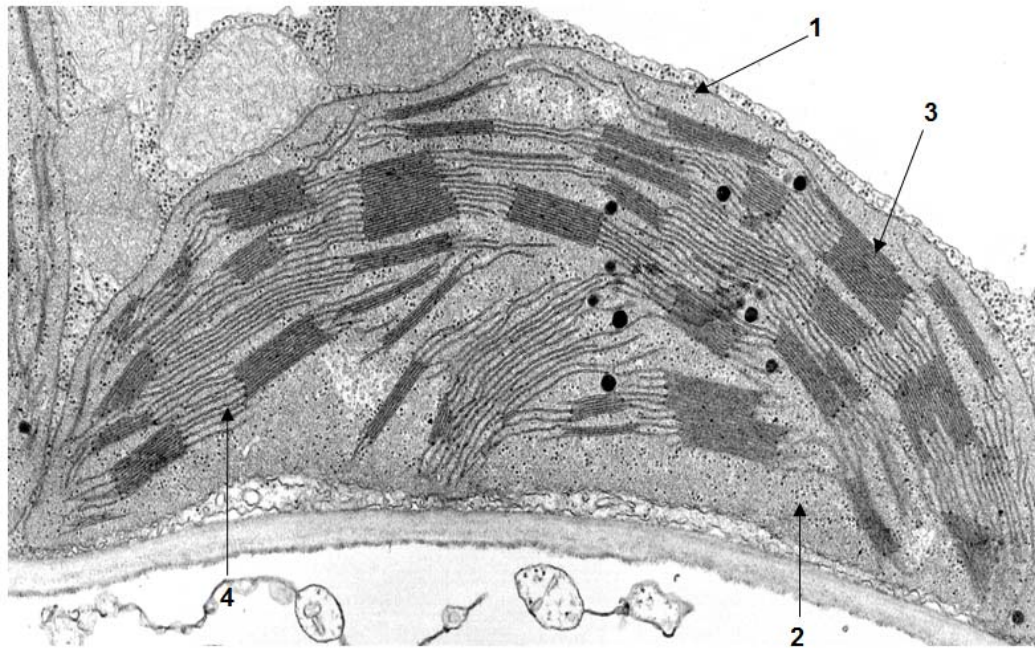
- 19** Four alleles determine coat colour in cats. The **C** allele codes for grey colour and is dominant over the **c^b** and **c^s** alleles. Both the **c^b** and **c^s** alleles are codominant and code for sepia colour. However, the **c^b** allele is also temperature sensitive and codes for black colour if during the fetal development stage, the cat is exposed to cool temperatures. Both the **c^b** and **c^s** alleles are dominant over the **c** allele which results in an albino.

Which of the following crosses could produce offsprings that have sepia colour with black patches?

- | | | | |
|------------|------------------|---|---------------------------------|
| I | grey colour cat | x | grey colour cat |
| II | sepia colour cat | x | sepia colour with black patches |
| III | grey colour cat | x | sepia colour cat |
| IV | sepia colour cat | x | sepia colour cat |

- A** I, II and III
- B** II and III
- C** II, III and IV
- D** All of the above

- 20 Which of the following is correctly associated with the function or characteristic of the labelled structure?

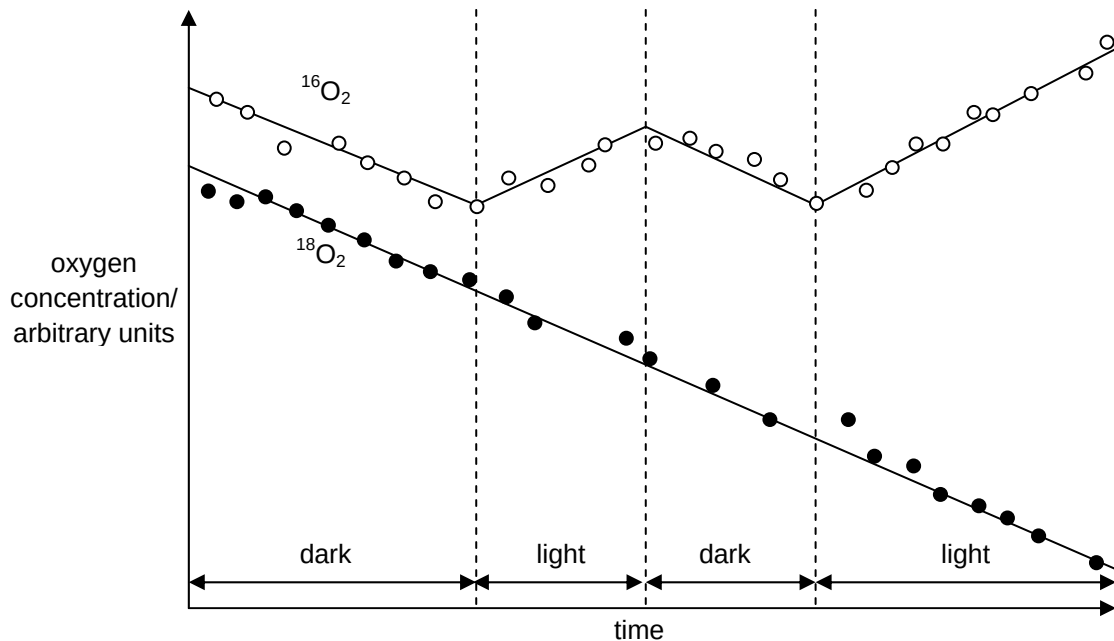


	help to maintain high H^+ concentration	made up of phospholipid	has stalked particles embedded in it	where ATP is synthesised
A	1 and 2	1, 2, 3 and 4	4	2
B	3 and 4	2 and 4	2 and 3	1 and 2
C	1 and 4	1, 2 and 3	1 and 4	2 and 4
D	3 and 4	1, 3 and 4	3 and 4	2

- 21 Isotopes of oxygen can be used to distinguish between oxygen absorbed by plants and oxygen evolved.

A mixture of oxygen isotopes $^{16}\text{O}_2$ and $^{18}\text{O}_2$ was supplied to a suspension of the unicellular alga *Chlorella* which had previously been exposed to $^{16}\text{O}_2$ only. During the following hour, changes in concentration of these gases in the suspension were measured in light and dark conditions.

The graph below shows the results.



What caused the concentration of $^{16}\text{O}_2$ to rise in light?

- A H_2^{18}O was being photolysed more rapidly than H_2^{16}O .
- B $^{16}\text{O}_2$ was being absorbed at different rates in light and dark.
- C $^{16}\text{O}_2$ was being produced in photosynthesis but was not being absorbed in respiration.
- D $^{16}\text{O}_2$ was being produced in photosynthesis faster than it was being absorbed in respiration.

22 Six tubes were set up as shown in the table below.

tube	Contents
1	Glucose + homogenized plant cells
2	Glucose + mitochondria
3	Glucose + cytoplasm lacking organelles
4	Pyruvate + homogenized animal cells
5	Pyruvate + mitochondria
6	Pyruvate + cytoplasm lacking organelles

After incubation, each sample was analysed to determine the presence of carbon dioxide and ethanol.

In which tubes are lactate most likely to be present?

- A** 1 and 3 only
- B** 2, 3, 5 and 6 only
- C** 4, 5, and 6 only
- D** 3 and 6 only

23 During aerobic respiration ATP can be formed by oxidative phosphorylation in the electron transport system and by glycolysis.

In the complete oxidation of two molecules of glucose, approximately what percentage of ATP is formed by oxidative phosphorylation?

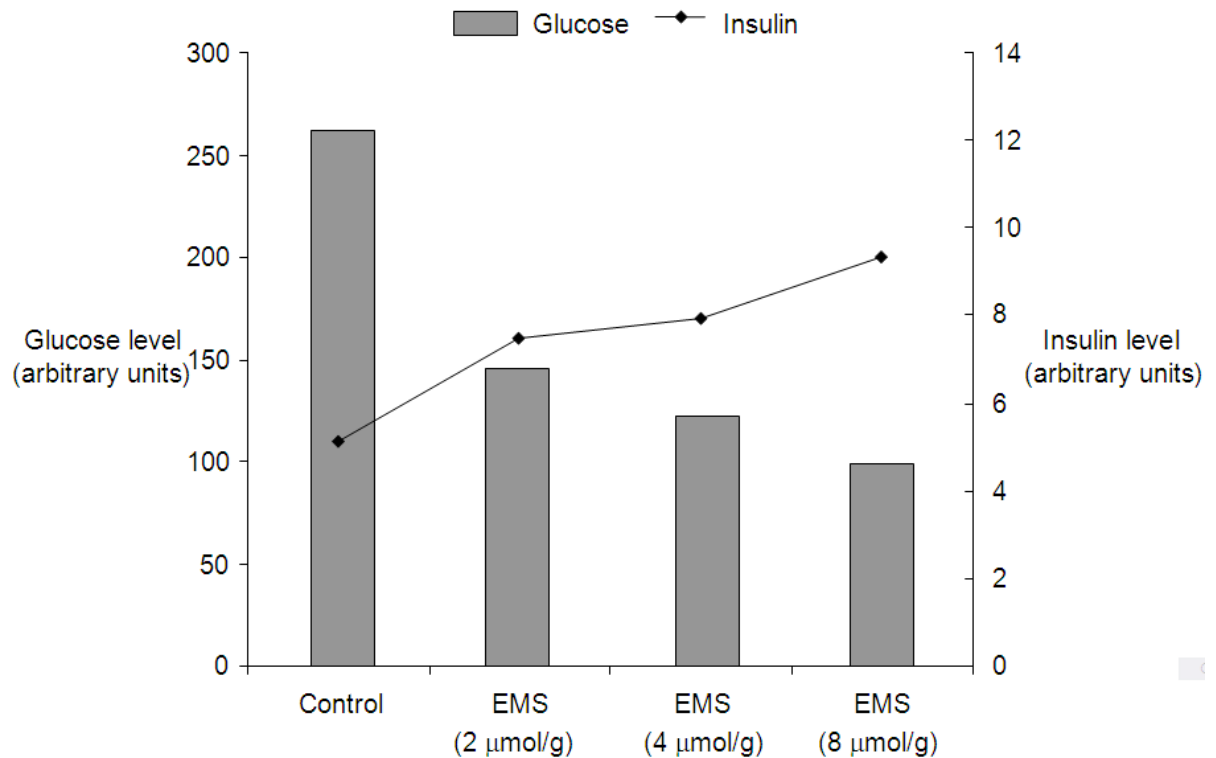
- A** 40%
- B** 45%
- C** 80%
- D** 90%

- 24** Steven has been unable to exert much energy. He goes to a doctor for help and is sent to the hospital for some tests. It was discovered that his mitochondria can use only fatty acids and amino acids for respiration, and his cells produce more lactate than normal.

Which of the following is the best explanation for his condition?

- A** His mitochondria lack the transport protein that moves pyruvate across the inner mitochondrial membrane.
- B** His cells cannot move NADH from glycolysis into the mitochondria.
- C** His cells contain an inhibitor that prevents oxygen use in his mitochondria.
- D** His cells have a defective electron transport chain, so glucose is reduced to lactate instead of to acetyl CoA.

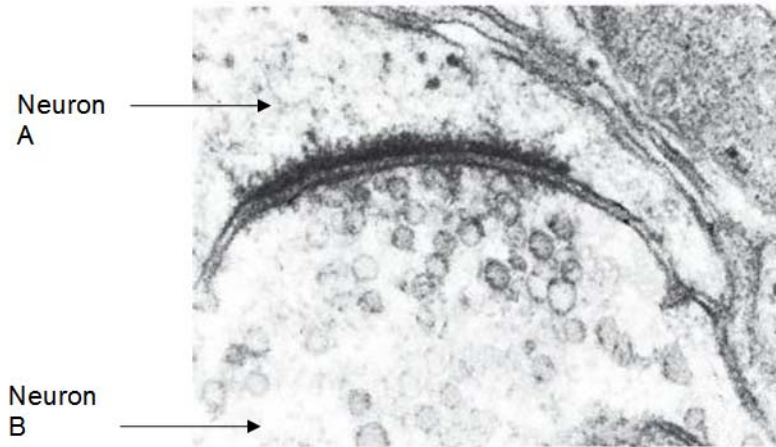
- 25** Succinic acid mono ethyl ester (EMS) is a drug which is used for the treatment of Type II diabetes. In a study, EMS was administered for 30 days to diabetic rats. The effect of EMS on blood glucose concentration and insulin were studied. The results are shown in the graph.



Which of the following explains why diabetics are encouraged to use EMS to treat their condition?

- A** The administration of EMS significantly reversed the high glucose levels in diabetic rats and is not dependent on the dosage of EMS.
- B** In diabetic rats treated with EMS, the level of plasma glucose was significantly increased whereas the level of plasma insulin was significantly decreased.
- C** EMS at a dose of 8 µmol/g showed a highly insignificant effect compared to 2 and 4 µmol/g.
- D** The administration of EMS to decrease the blood glucose concentration is an essential trigger for the body to revert to its normal homeostasis.

26



The micrograph above shows a synapse between two neurons. Which of the following statements about synaptic transmission is correct?

- A Neuron B has received the synaptic vesicles from Neuron A.
- B The membrane of Neuron A has many neurotransmitter channels.
- C Calcium influx is triggered by depolarization that is produced by the arrival of an action potential in Neuron B.
- D Neurotransmitters triggers voltage-gated sodium ion channels in Neuron A.

27 Which of the following describes the state of a non-propagating (resting) mammalian axon?

	Concentration in axoplasm		Relative axon permeability	Condition of cation pumps in axon membrane
	Na ⁺	K ⁺		
A	High	Low	Low	Active
B	High	Low	High	Inactive
C	Low	High	High	Inactive
D	Low	High	Low	Active

28 Which of the following process(s) leads to signal amplification in the cell?

- I** Binding of multiple signal molecules to the membrane bound receptor
- II** Phosphorylation of the first kinase of a cascade
- III** Branching of different signaling pathways that produces multiple cellular responses
- IV** Activation of numerous intracellular receptors that acts as transcription factors in gene expression

- A** II only
- B** II and III
- C** I and IV
- D** All of the above

- 29** A scientist is investigating the effects of Poison T on the cell signaling pathway of glucagon. It is found that Poison T diminishes the effect of glucagon.

The table below shows the different components of the cell signaling pathway of glucagon and their statuses.

Component	Status
G Protein Coupled Receptor (GPCR)	Configuration can be changed
G protein	Can exchange GDP for GTP
cAMP levels	low
Protein Kinase A	Inactivated

Which of the following statements can be concluded from the results?

- 1** Poison T interferes with reception.
 - 2** Poison T prevents G protein from hydrolysing GTP.
 - 3** Poison T inactivates the enzyme adenylate cyclase.
 - 4** Poison T stops the relaying of message down the phosphorylation
-
- A** 1 and 2
 - B** 2 and 3
 - C** 2 and 4
 - D** 3 and 4

- 30** Cell signalling is mediated by five mutant subtypes of membrane growth factor receptor. These five mutants are structurally similar to insulin receptors. Various functional abilities are tested and the results are summarised in the table.

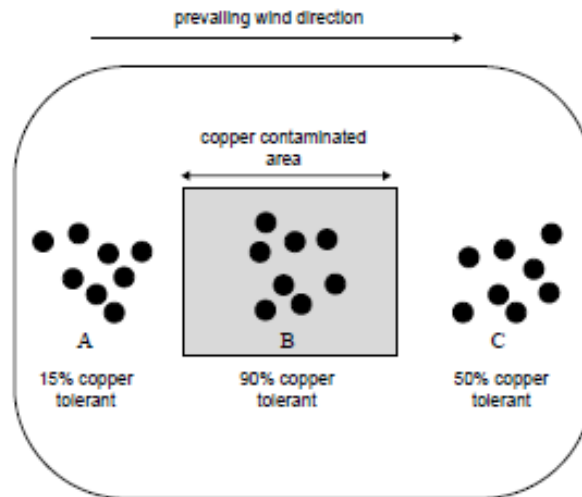
(The signs “+” and “-” indicate respectively the presence and absence of the indicated functions.)

Receptor subtype	Functions				
	Binding of growth factor	Conformational changes	Hydrolysis of PIP ₂	Auto-phosphorylation	Protein kinase activity
WT	+	+	+	+	+
ATP	+	-	-	-	-
CT	+	-	-	-	-
TM	+	-	-	-	-
NOKI	+	-	+	+	-

Which of the following conclusions is **NOT** likely?

- A** **ATP**, **CT** and **TM** receptors are deficient in tyrosine kinase activity.
- B** Conformational changes in the receptor are necessary for autophosphorylation.
- C** **WT** receptors mediate phospholipase C activation.
- D** Conformational changes of the receptor are necessary for protein kinase activation.

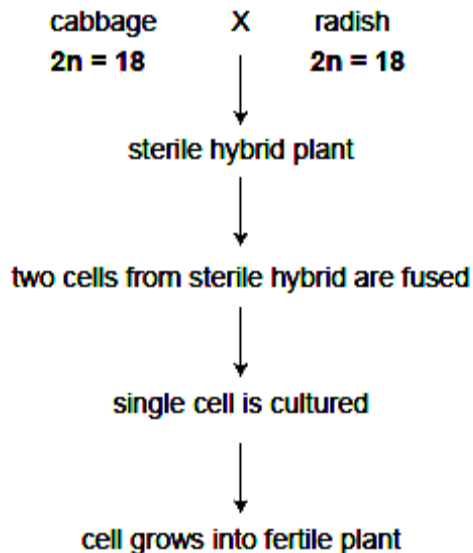
- 31** The figure below shows an aerial view of a region which included a section of copper contaminated soil (shaded grey on the diagram). The black circles represent populations of grass growing in areas A, B and C. The figure also includes the percentage of copper resistant plants in areas A, B and C. Tolerance to copper in the soil is genetically determined.



It is reasonable to conclude from the information that _____ .

- A** all plants in the copper contaminated area would be homozygous at the gene locus for copper tolerance
- B** high levels of copper tolerance would be a selective advantage for all plants in areas A, B and C
- C** the difference in copper tolerance between populations in areas A and C is the result of mutation
- D** gene flow is occurring between populations in areas B and C

- 32** Cabbage, *Brassica oleracea*, and radish, *Raphanus sativus*, are both members of the Brassicaceae family. The cells of each type of plant have 18 chromosomes. When these plants are crossed, a sterile hybrid is produced. Two cells from a sterile hybrid plant are fused and the single cell that is formed is cultured. The flowchart shows the steps involved.



Which of the following can be concluded from the flowchart?

- A** Cabbage and radish have identical chromosomes as they both belong to the Brassicaceae family.
 - B** Gametes from the fertile plant will be identical to those of the cabbage or the radish.
 - C** The fused cell has 18 pairs of homologous chromosomes and the sterile hybrid has nine pairs of homologous chromosomes.
 - D** The eventual fertile plant is considered an allopolyploid.
- 33** Which one of the following statements is related to the principle of adaptive radiation in evolution?
- A** Unrelated organisms living in similar environments may tend to resemble each other.
 - B** Related organisms living in different environments show modifications of an underlying unity of pattern.
 - C** Selection in a constant environment tends to maintain the most frequently occurring variety of a species.
 - D** Species evolve by selection altering the gene frequencies in interbreeding populations.

- 34** A species of seagulls exists as a large number of different populations on separate islands of the Azores archipelago. Each population contains 100 individuals.

In the original populations, the black-colored seagull is due to a dominant allele while white-colored individuals are due to recessive allele. In each population, 81% individuals are white-colored.

During a storm, two males and two females were blown 300 miles to the West, to an isolated island uninhabited by seagulls (Survivor Island). They reproduced and founded a new population. They could not disperse back to the original habitat, nor can any new seagulls cross to the new island. By chance, all four original colonists were white-colored seagulls.

Over the course of the next 200 years, these Survivor Island seagulls evolved mating displays that differ from those of the parent population. When television producers introduce the original Azores archipelago seagulls to the island, the two kinds of seagulls do not interbreed.

Which of the following options is **CORRECT** regarding the paragraphs above?

	Frequency of heterozygous black-colored in each original population	New frequency of the recessive allele after the storm	Type of isolation mechanism keeps the two species of gulls from interbreeding
A	0.00	0.81	Behavioral Isolation
B	0.90	1.00	Mechanical Isolation
C	0.18	1.00	Behavioral Isolation
D	0.81	0.00	Mechanical Isolation

- 35** Which of the following poses the greatest limitation of the Human Genome Project in disease analysis?

- A** It is not possible to store the genomic information of all individuals.
- B** There is too large a genetic difference between individuals to determine the loci of genes of interest.
- C** Too few model organisms exist for comparative genomic studies to take place.
- D** Environmental factors and gene interactions in disease are not taken into account.

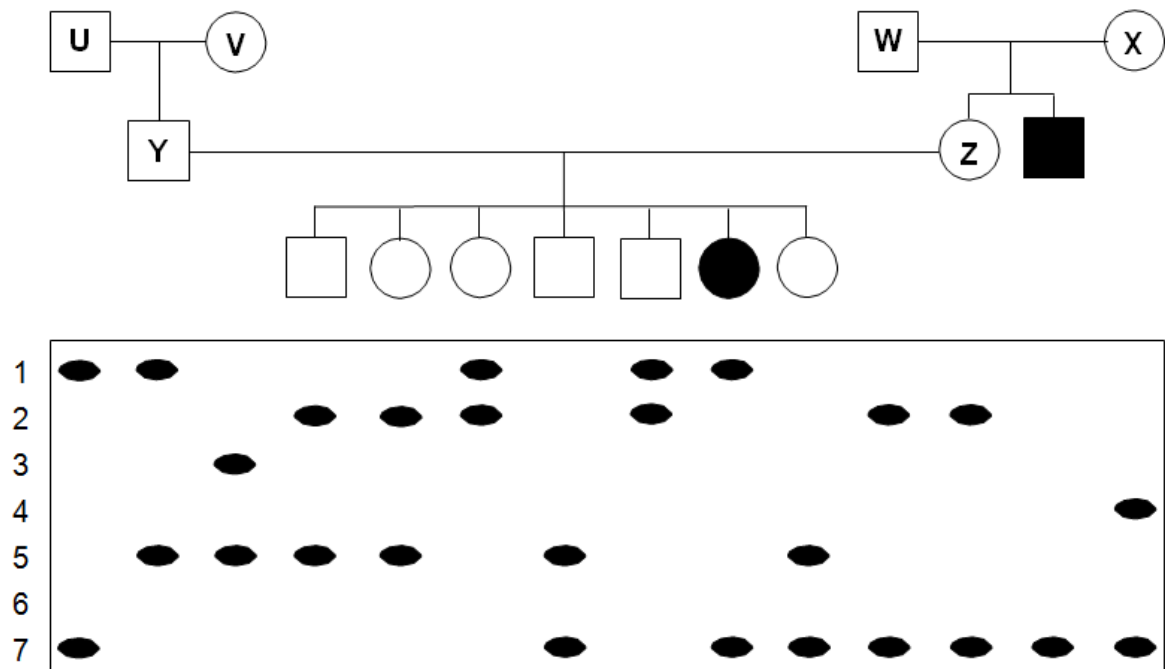
- 36** A sample of DNA was treated separately with two different restriction enzymes. The products of digestion was then run on a gel and stained with ethidium bromide.

Which of the following statement(s) **CANNOT** account for the difference in the *number* of bands seen in the gel?

- I The two enzymes have active sites that are of different shapes.
- II One of the enzymes produces blunt ends and the other sticky ends.
- III The position of the restriction sites for each restriction enzyme differs
- IV Ethidium bromide binds specifically to only certain DNA fragments.

- A II, III and IV only
- B I and III only
- C II and IV only
- D II only

- 37 A DNA polymorphism with 7 alleles,(1-7), is linked to the gene causing Disease H (an autosomal recessive disease).



In which individual has crossing over taken place between the DNA polymorphism and the gene loci?

- A U only
 - B V only
 - C W and X only
 - D Y and Z only
- 38 A patient underwent gene therapy using an adenoviral mediated delivery system. The gene therapy failed and you were asked to find out the cause of the failure. Which of the following could **NOT** be the explanation behind the failure in the gene therapy?
- A Failure of the expressed protein to be folded into the correct conformation during post-translational modifications.
 - B The protein was not expressed in adequate amount.
 - C The target gene failed to integrate next to a suitable promoter in the host chromosome.
 - D Rejection and destruction of the vector by the host immune system.

- 39** A patient suffers from adenosine deaminase (ADA) deficiency, an autosomal recessive immune deficiency in which bone marrow lymphoblasts cannot replicate to generate immunocompetent lymphocytes. The treatment option that would permanently cure the patient is _____.
- A** germ-line gene therapy to replace one ADA gene copy
 - B** germ-line gene therapy to replace both ADA gene copies
 - C** somatic cell gene therapy to replace both ADA gene copies in circulating lymphocytes
 - D** somatic cell gene therapy to replace one ADA gene copy in bone marrow lymphoblasts
- 40** Which of the following statements can best be considered to be a social implication of genetically modified (GM) crop plants?
- A** There is evidence that genes for herbicide resistance have spread from (GM) oilseed rape to nearby weed species although there is no evidence that these hybrids have persisted in the environment.
 - B** In some studies, the number of beneficial insects has been reduced in areas where insect resistant GM Bt corn has been grown, however in other studies, there has been no effect or the number of beneficial insects has increased.
 - C** Pressure placed by people on a multinational GM seed company has caused them to pledge that they will never develop a 'suicide' gene that prevents farmers from growing seed collected from last year's crop.
 - D** In some countries, government regulations have been put in place to prevent farmers planting GM crops close to non-GM and organic crops to prevent GM pollen from reaching the crops.

END OF PAPER

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2013 Y6 Preliminary Exam H2**MCQ Answer Scheme**

1	A	21	D
2	B	22	D
3	C	23	D
4	D	24	A
5	B	25	D
6	C	26	C
7	A	27	D
8	D	28	A
9	B	29	D
10	D	30	B
11	A	31	D
12	A	32	D
13	D	33	B
14	B	34	C
15	D	35	D
16	C	36	A
17	C	37	A
18	C	38	C
19	D	39	D
20	D	40	C